e.g:- $(25.625)_{10}$

2 | 25		$0.625 \times 2 = 1.250 \rightarrow 1$
2 | 12 → 1		$1.250 \times 2 = 2.500 \rightarrow 2$
2 | 6 → 0		$2.500 \times 2 = 5.000 \rightarrow 5$
2 | 3 → 0		
2 | 1 → 1		$(0.625)_{10} = (0.125)_2$
0 → 1		

$(25)_{10} = (11001)_2$

$\therefore (25.625)_{10} = (11001.125)_2$

e.g:- $(25.625)_{10}$

2 | 25		$0.625 \times 2 = 1.250 \rightarrow 1$ ↓
2 | 12 → 1	↑	$0.250 \times 2 = 0.500 \rightarrow 0$ ↓
2 | 6 → 0	↑	$0.500 \times 2 = 1.000 \rightarrow 1$
2 | 3 → 0	↑	
2 | 1 → 1	↑	$(0.625)_{10} = (0.101)$
0 → 1		

$(25)_{10} = (11001)_2$

$\therefore (25.625)_{10} = (11001.101)_2$

2] Decimal to octal conversion :

Step :- i] Divide decimal by 8
ii] Record remainder
iii] Divide quotient again by 8
iv] Repeat the process until equotient is 0
v] Read the remainder in reverse order (bottom to top)

eg:- $(75)_{10} = (?)_8$

8 | 75		$(75)_{10} = (113)_8$
8 | 9 → 3	↑	
8 | 1 → 1	↑	
0 → 1		

$(851)_{10} = (?)_8$

8 | 851		$(851)_{10} = (1523)_8$
8 | 106 → 3		
8 | 13 → 2		
8 | 1 → 5		
0 → 1		

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Q $(1011.010)_2$
$-(0110.101)_2$

Borrow

0100.101

Q 10101100
-00111000

Borrow

01110100

$(01110100)_2$

" Binary multiplication

A	B	Product
0	0	0
0	1	0
1	0	0
1	1	1

Q $(9)_{10} \times (11)_{10}$

1001
$\times 1011$
1001 ? (1001)
$\times$

Q $(9)_{10} \times (11)_{10}$

1001
$\times 1011$
1001
$1001x$
$+ 1001xxx$
carry
1100011

$(1100011)_2 = (99)_{10}$

Q 10110
$\times 101$
10110
$+ 10110xx$
carry
1101110

$(1101110)_2$

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Hw :- 9-8 , $1^{s}C$ and $2^{s}C$

1] 1^{s} complement subtraction method

$(9)_{10} = (1001)_2$
$(8)_{10} = (1000)_2$

$1^{s}C$ of $(8)_{10}$ = 0111 = -8

$$\begin{array}{rrl} 9 & = & 1001 \\ +(-8) & + & 0111 \\ & & \hline & & 111 \quad \text{carry} \\ & & \boxed{1}0000 \\ & & \downarrow \\ & & 1 \end{array}$$

Carry is generated.

∴ Carry will be added to the result true form

$= 0000 + 1$

$= 0001 = (+1)_{10}$

2] 2^{s} complement subtraction method

$(9)_{10} = (1001)_2$
$(8)_{10} = (1000)_2$

$2^{s}C$ of $(8)_{10}$ = $1^{s}C$ of 8 + 1
= 0111 + 1
= 1000

$$\begin{array}{r} 0111 \\ \quad\quad 1 \\ \hline 1\,11 \\ 1000 \end{array}$$

$$\begin{array}{rrl} 9 & = & 1001 \\ +(-8) & & +1000 \\ & & \hline & & \boxed{1}0001 \\ & & \downarrow \\ & & 1 \end{array} \qquad \begin{array}{r} 1001 \\ 1000 \\ \hline 1 \\ \hline 10001 \end{array}$$

Carry is generated

∴ discard the carry, also the result will be in true form and positive.

∴ answer = $(0001)_2 = (1)_{10}$

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Theorem 5 :- $A + \bar{A}B = A + A\bar{B} + \bar{A}B$

$\therefore \boxed{A + \bar{A}B = A + B}$

6] DeMorgans

1st law i] Sum of complement is equal to product of each complement

$\boxed{\overline{A+B} = \bar{A} \cdot \bar{B}}$

2nd law ii] Product of complement is equal to sum of each complement

$\boxed{\overline{A \cdot B} = \bar{A} + \bar{B}}$

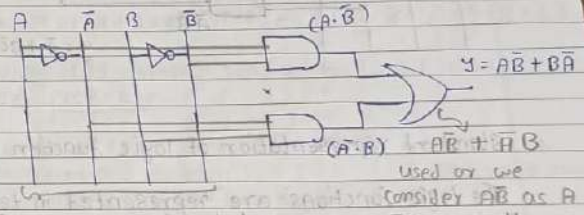
7] Complement :- $\boxed{A + \bar{A} = 1}$ $\boxed{A \cdot \bar{A} = 0}$

" Circuit realization :-

realize the following boolean equation

i] $Y = A\bar{B} + \bar{A}B$ ii] $Y = \bar{A} + B\bar{C} + \bar{A}B$

i] $Y = A\bar{B} + \bar{A}B$



$A\bar{B} + \bar{A}B$ used or we consider $A\bar{B}$ as A $\bar{A}B$ as B thus OR(+) gate used.

jis gate ki jo value hai vo property join kara

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